

FUNDAMENTALS OF COMPUTER NETWORK

COMPUTER NETWORK

■ DEFINITION:

A computer network is defined as the interconnection of two or more computers. It is done to enable the computers to communicate and sharing.

■ A computer network is a system of interconnected computers and peripheral devices.

■ For example, it may connect computers, printers, scanners and cameras.re available resources.

APPLICATIONS:

■ Why do we need computer network:

- Sharing of resources such as printers
- Sharing of expensive software's and database
- Communication from one computer to another computer
- Exchange of data and information among users via network
- Sharing of information over geographically wide areas.

COMPONENTS OF COMPUTER NETWORK :

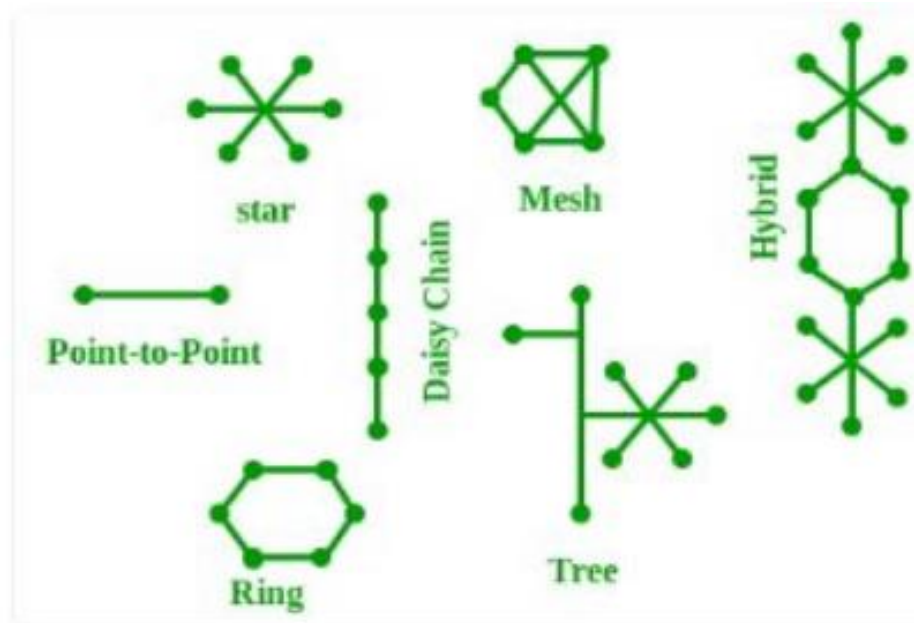
- Two or more computers
- Cables as links between the computers
- A network interfacing card(NIC) on each
- computer
- Switches
- Software called operating system (OS)

BEFORE WE START!! TERMINOLOGY:

- **Open system:** A system which is connected to the network and is ready for communication.
- **Closed system:** A system which is not connected to the network and can't be communicated with.
- **Computer Network:** It is the interconnection of multiple devices, generally termed as Hosts connected using multiple paths for the purpose of sending/receiving data or media
- There are also multiple devices or mediums which helps in the communication between two different devices which are known as Network devices. Ex: Router, Switch, Hub, Bridge



■ The layout pattern using which devices are interconnected is called as network topology. Such as Bus, Star, Mesh, Ring, Daisy chain



■ **OSI:** OSI stands for Open Systems Interconnection. It is a reference model that specifies standards for communications protocols and also the functionalities of each layer.

■ **Protocol:** Protocol is the set of rules or algorithms which define the way how two entities can communicate across the network and there exists different protocol defined at each layer of OSI model. Few of such protocols are TCP, IP, UDP, ARP, DHCP, FTP and so on.

■ UNIQUE IDENTIFIERS OF NETWORK:

Host name: Each device in the network is associated with a unique device name known as Hostname. Type “hostname”

in the command prompt and press 'Enter', this displays the hostname of your machine

■ IP Address (Internet Protocol address):

Also known as Logical Address, is the network address of the system across the network. To identify each device in the world-wide web, Internet Assigned Numbers Authority (IANA) assigns IPV4 (Version 4) address as unique identifier for each device on the Internet. Length of the IP address is: 32-bits. (Hence we have 232 ip addresses available.) Type "ipconfig" in the command prompt and press 'Enter', this gives us the IP address of the device.

■ MAC Address (Media Access Control address):

Also known as physical address, is the unique identifier of each host and is associated with the NIC (Network Interface Card). MAC address is assigned to the NIC at the time of manufacturing. Length of the MAC address is: 12-digit/ 6 bytes/ 48 bits Type "ipconfig/all" in the command prompt and press 'Enter', this gives us the MAC address.

■ Port:

Port can be referred as logical channel through which data can be sent/received to an application. Any host may have multiple applications running, and each of this application is

identified using the port number on which they are running on. Port number is a 16-bit integer , hence we have 2^{16} ports available which are categorized as shown below:

PORT TYPES	RANGE
Well known Ports	0 – 1023
Registered Ports	1024 – 49151
Ephemeral Ports	49152 – 65535

■ Number of ports: 65,536

Range: 0 – 65535

Type “netstat -a” in the command prompt and press ‘Enter’, this lists all the ports being used.

■ Socket:

The unique combination of IP address and Port number together are termed as Socket.

■ DNS Server:

DNS stands for Domain Name system. DNS is basically a server which translate web addresses or URL (ex: www.google.com) into their corresponding IP addresses. We don't have to remember all the IP addresses of each and every website. The command ‘nslookup’ gives you the IP

address of the domain you are looking for. This also provides the information of our DNS Server.

LETS START!!

Computer network

- A computer network is a system of interconnected computers and peripheral devices.
- For example, it may connect computers , printers, scanners and cameras.

Communications

■ Communications is about the transfer of information from a sender, across a distance, to a receiver.

■ Communication is an act of transmitting messages.

Computer network communication

■ We transmit information or data by using two types of signals, namely analog and digital.

■ Computers communicate with digital signals.

■ The older forms of communications technology, such as telephones and radios, use analog signals.

NETWORK BENEFITS

■ The network provided to the users can be divided into two categories: ▪ Sharing ▪ Connectivity

■ Types of resources are:

- Hardware: A network allows users to share many hardware devices such as printers , modems, fax machines, CD ROM, players, etc.
- Software: sharing software resources reduces the cost of software installation, saves space on hard disk.

BENEFITS

Main benefits of networks include:

- File sharing – you can easily share data between different users, or access it remotely if you keep it on other connected devices.
- Resource sharing – using network-connected peripheral devices like printers, scanners and copiers, or sharing software between multiple users, saves money.
- Sharing a single internet connection – it is cost-efficient and can help protect your systems if you properly secure the network.
- Increasing storage capacity – you can access files and multimedia, such as images and music, which you store remotely on other machines or network-attached storage devices. Improved Communication

■ Networking computers can also help you improve communication, so that:

- staff, suppliers and customers can share information and get in touch more easily
 - your business can become more efficient – eg networked access to a common database can avoid the same data being keyed multiple times, saving time and preventing errors
 - staff can deal with queries and deliver a better standard of service as a result of sharing customer data
- Cost benefits of computer networking

■ Storing information in one centralized database can also help you reduce costs and drive efficiency.

■ For example: staff can deal with more customers in less time since they have shared access to customer and product databases you can centralize network administration, meaning less IT support is required you can cut costs through sharing of peripherals and internet access

■ You can reduce errors and improve consistency by having all staff work from a single source of information. This way, you can make standard versions of manuals and directories available to them, and back up data from a single point on a scheduled basis, ensuring consistency.

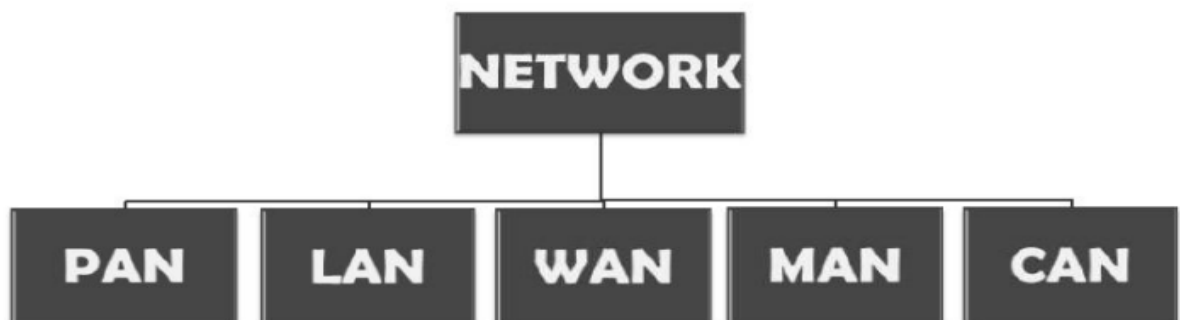
OTHER BENEFITS OF COMPUTER NETWORK

- Increased speed
- Reduced cost
- Improved security
- Centralized software managements
- Electronic mail
- Flexible access

DISDAVATAGES OF NETWORKS

- High cost of installation
- Requires time for administration
- Failure of server
- Cable faults

CLASSIFICATION OF AREA BY THEIR GEOGRAPHY

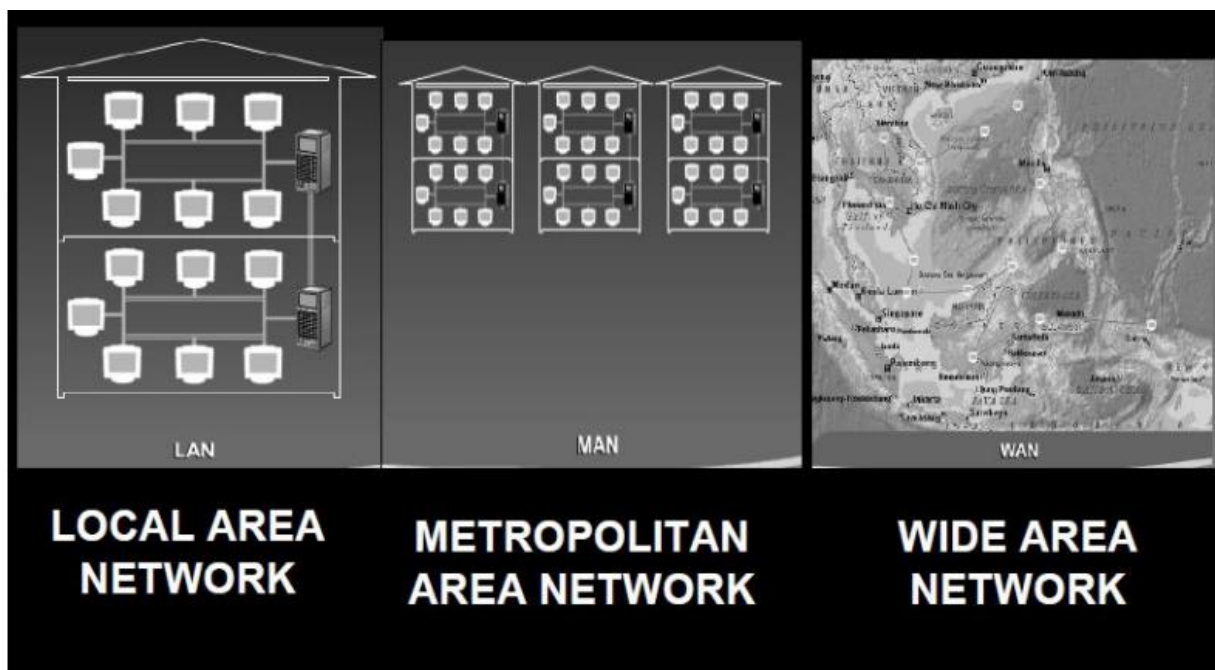


PERSONAL AREA NETWORK (PAN):

- A personal area network is a computer network organized around an individual person.
- It generally consists of a mobile computer, a cell phone or personal digital assistant.
- PAN enables the communication among these devices.
- It can also be used for communication among personal devices themselves for connecting to a digital level network and internet.

The PANs can be constructed using wireless or cables.

TYPES OF NETWORKS:



LOCAL AREA NETWORK(LAN):

- LAN is a network which is designed to operate over a small physical area such as an office, factory or a group of buildings.
- LAN's are easy to design and troubleshoot
- Exchange of information and sharing of resources becomes easy because of LAN.
- In LAN all machines are connected to a single cable.
- Different types of topologies such as star, tree, bus, ring, etc. Can be used
- It is usually a privately owned network.
- Smallest network compared to the other two networks.
- The simplest form of LAN is to connect two computers together.
- LAN is operated within a limited physical area, such as at home, school, a single building or several buildings.
- A network which consists of less than 500 interconnected devices across several buildings, is still recognized as a LAN.

METROPOLITAN AREA NETWORK (MAN)

- 'Metropolitan' - describes important cities like New Delhi, Mumbai, Kolkata, bangaluru, Chennai etc.

- Companies that have several branches within the Mumbai city such as banks, use a MAN.
- Can be a collection of several LANs within the same city.

MAN can be defined as a group of computers and network devices connected together within a large physical area.

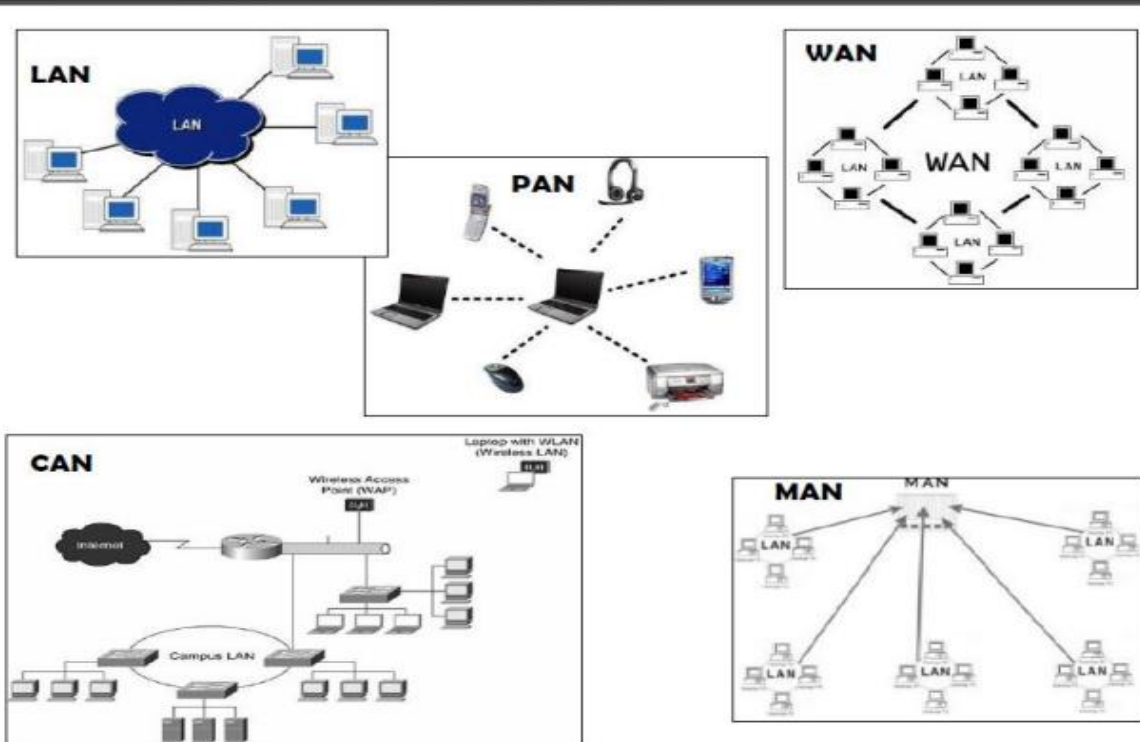
WIDE AREA NETWORK(WAN)

- When network spans over a large distance or when the computers to be connected to each other are at widely separated locations a local area network cannot be used.
- A wide area network(WAN) is installed. The communication between different users of WAN is established using leased telephone lines, satellite links and similar channels.
- It is cheaper and more efficient to use the phone network for the link.
- Most WAN networks are used to transfer large blocks of data between its users.
- The largest network of all network types.
- Internet is the largest WAN in the world.
- WAN generally covers large distances such as states, countries or continents.

- **EXAMPLE:** Local banks have always maintained their business online by connecting all computers of their branches in the countries. International banks also use WAN to connect their computers all over the world.
- WAN is a group of MANs or LANs or the mixture of both networks.

CAMPUS AREA NETWORK(CAN)

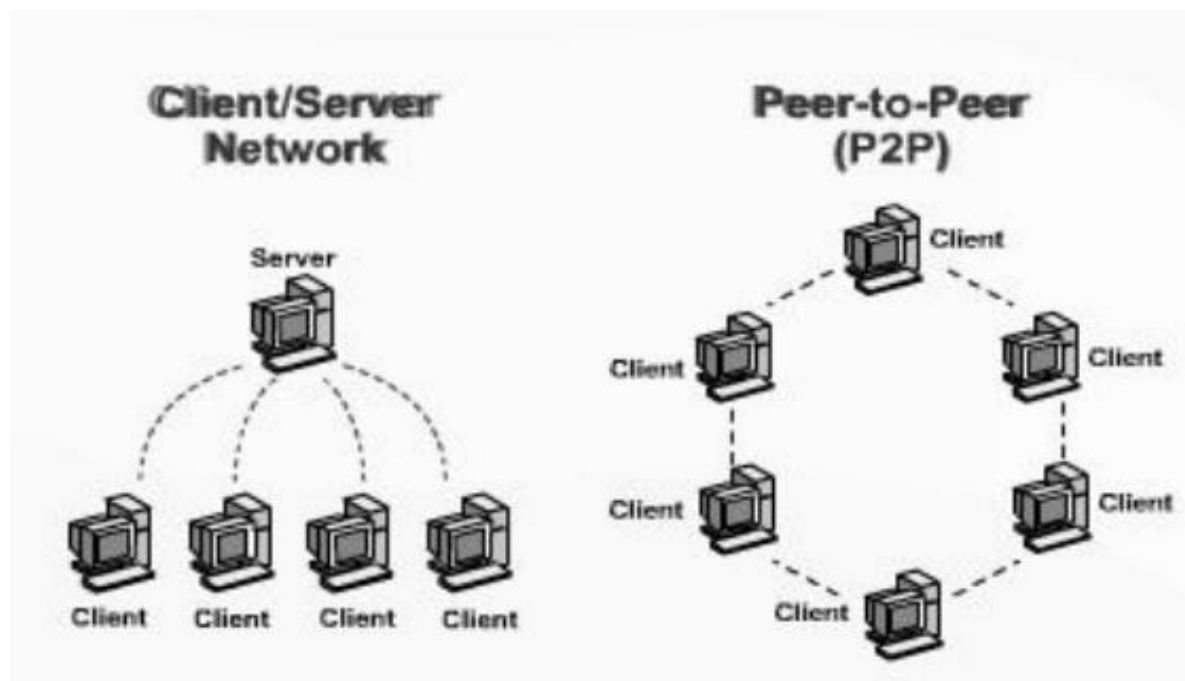
- The campus area network is made up of an interconnection of LAN with limited geographical area.
- Network equipments such as switches, routers and the transmission media i.e. optical fiber etc. are almost entirely owned by the campus owner.



Comparison of LAN, MAN, WAN :

CRITERIA	LAN	MAN	WAN
Cost	Low	High	Higher
Network Size	Small	Larger	Largest
Speed	Fastest	Slower	Slowest
Transmission media type	Twisted-pair	Twisted-pair and fibre-optic cables	Fiber optic, radio wave and satellite
Number of computers	Smallest	Large	Largest

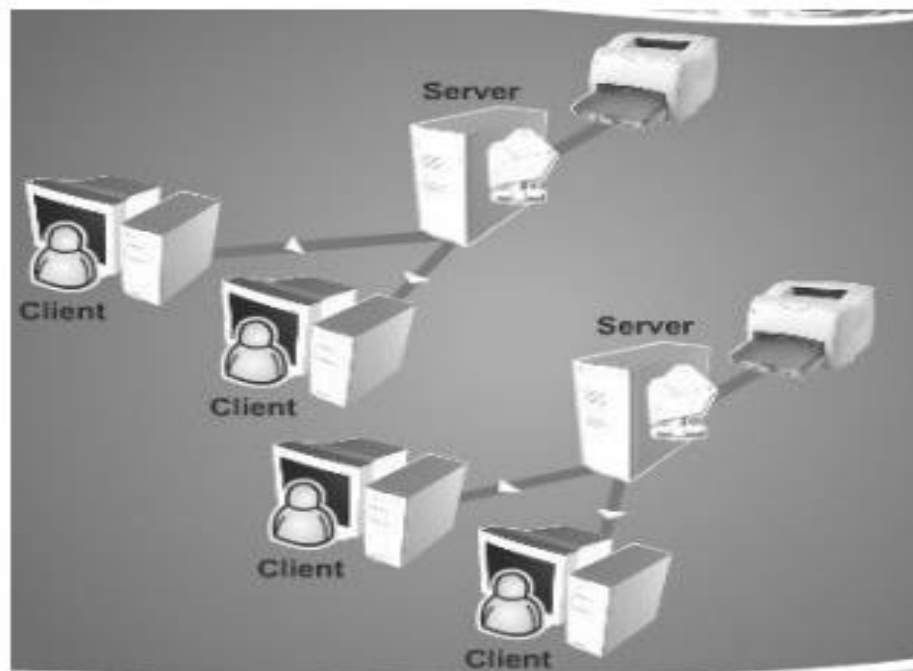
NETWORK ARCHITECTURE:



- how a computer network is configured and what strategies are being used.
- the functions of the networks.
- Also known as network model or network design.

CLIENT/SERVER NETWORK

- A client/server network is a network in which the shared files and applications are stored in the server but network users (clients) can still store files on their individual PCs.
- A server is a computer that shares Information and resources with other computers on a network.
- A client is a computer which requests services or files from a server computer.



Peer-to-peer or P2P

- It is a network with all the nodes acting as both servers and clients.
- A PC can access files located on another PC and can also provide files to other PCs.
- All computers in the peer-to-peer network has equal responsibilities and capabilities to use the resources available on the network.
- With peer-to-peer network, no server is needed; each computer in the network is called a peer.

